WEEK 5 : ADVANCED MACHINE LEARNING

Performance Comparison Report

1. Problem Statement

The goal of this project is to predict residential house prices based on structural features and locational attribute of the property (proximity to roads, number of bedrooms, availability of parking area .e.t.c)

2.Dataset Description

The Dataset (“housing.csv”) contains 545 records with 13 columns: Price(target variable), area, bedrooms, bathrooms, stories, parking, airconditioning, prefarea, mainroad, guestrooms, basement, hotwaterheating, furnishingstatus. No missing values or duplicate rows were present.

Pre-processing steps carried out before modelling:

- Binary columns were encoded as 0/1
- Area and Price were log-transformed to reduce skewness
- Furnishing Status was encoded ordinally
- Features were standardized using StandardScaler for Linear Model, tree-based models were trained on unscaled features
- random_state = 42 for consistent reproducible comparison across all models

3. Models Used

- Baseline: Linear Regression
- Decision Tree Regressor (max_depth = 5)
- Random Forest Regressor (default and GridSearchCV-tuned)
- Gradient Boosting Regressor (default and RandomizedSearchCV-tuned)

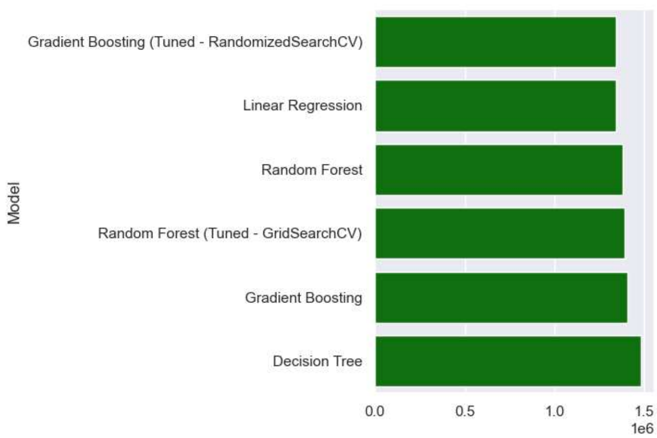
4. Evaluation Metrics

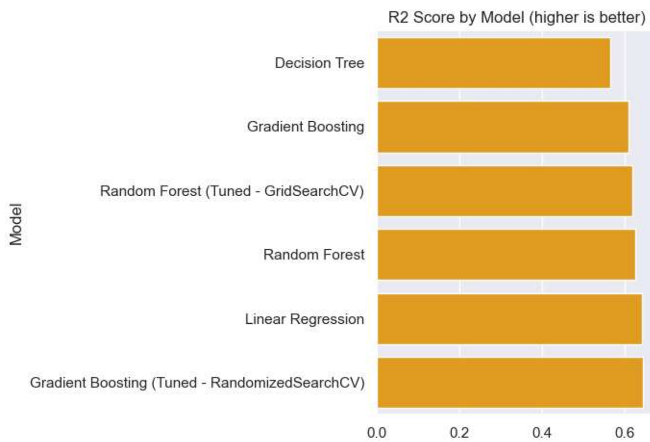
- Mean Absolute Error (MAE)
- Mean Squared Error(MSE)
- Root Mean Squared Error(RSME)
- R2 Score(R2)
  MAE,RSME & MSE are reported in currency

5.Performance Comparison

MODEL	MAE	MSE	RSME	R2

Linear Regression	1002010.28	1806438478634.45	1344038.12	0.6426
Decision Tree	1041559.89	2197657797710.50	1482449.93	0.5652
Random Forest	958271.18	1898734415467.04	1377945.72	0.62435
Gradient Boosting	985965.66	1977062622978.10	1406080.59	0.6089
Random Forest (Tuned)	966125.9309	1927030087582.1777	1388175.09255	0.6188
Gradient Boosting (Tuned)	954460.9667	1798071521030.4753	1340921.8922	0.6443





6. KEY FINDINGS

- Gradient Boosting Regressor (default and RandomizedSearchCV-tuned) was the best performing model {MAE': 954460.9666566127, 'MSE': 1798071521030.4753, 'RMSE': np.float64(1340921.892218363) 'R2': 0.6442681705085742}
- Tuning had opposite effects on the two enemble models(Random Forest and Gradient Boosting). It increased R2 on Dradient Boosting (0.6089 to 0.6443) but slightly reduced for Random Forest (0.6244 to 0.6188). This shows that hyperparameter tuning doesn't uniformly help all algorithms and it needs to be evaluated per-model
- The Single Decision Tree was the weakest Model(R2 = 0.5652), with 545 records to parse, a lone tree tends to overfit and learn the training data rather than generalizing. Ensemble models fix this by either averaging numerous trees (RandomForest) or sequencing them (GradientBoosting)
- LinearRegression performed exceptionally well (R2 = 0.6426). This shows that the relationship between the selected features and the target variable (price) is highly linear once log-transforms and ordinal encodings are applied.